A. Adjacency Matrix Representation

time limit per test: 1 second
memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

You are given a **directed weighted** graph with ***N*** nodes and ***M*** edges. The nodes are numbered from 1 to N. Each edge represents a direct connection between two nodes. There is no self loop or multi edge.

Input

The first line contains two integers N and M $(1 \leq N \leq 100, 0 \leq M \leq \frac{N(N-1)}{2})$ — the number of vertices and the total number of edges.

The next M lines will contain three integers $u_i, v_i, w_i (1 \leq u_i, v_i \leq N, 1 \leq w_i \leq 1000)$ — denoting there is an edge from node u_i to v_i with cost w_i.

Output

The output consists of an $N \times N$ adjacency matrix representing the directed weighted graph. Each row corresponds to a node, and each column represents its directed edges to other nodes. The value at position (i, j) denotes the weight of the edge from node i to node j. If there is no edge, the value is 0.

Examples

input

```
6 7
1 5 6
6 3 5
1 3 9
3 4 7
4 6 1
5 6 8
6 1 6
```

output

```
0 0 9 0 6 0
0 0 0 0 0 0
0 0 0 7 0 0
0 0 0 0 0 1
0 0 0 0 0 8
6 0 5 0 0 0
```

input

```
4 3
1 3 8
3 2 5
1 4 2
```

output

```
0 0 8 2
0 0 0 0
0 5 0 0
0 0 0 0
```